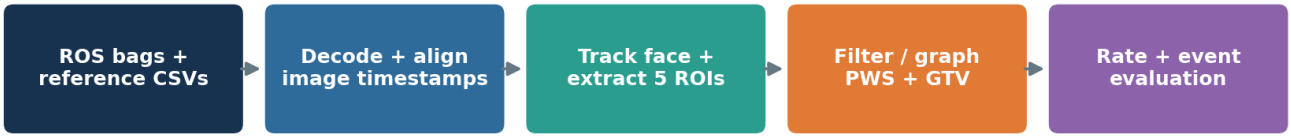
PROJECT REPORT | 31 OCTOBER 2026

Thermal Respiration Monitoring

Reconstruction, signal processing, and baseline evaluation on the L-CAS dataset

5 recordings / subjects	**14,154** thermal frames
20 owner-reviewed sample frames	**115** visible reference boxes

Main finding: the simple bandpass peak/valley detector outperformed the tested TCN event detector. A mouth-region spectral baseline also had lower error than the nose baseline in this five-subject exploratory comparison. Neither result is independently validated.



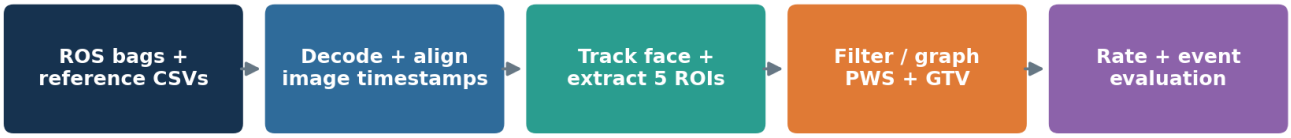
Owner-confirmed event map: 0 = no event, 1 = inhalation, 2 = exhalation (not author-verified)

This report separates verified data-processing work from exploratory model results. The face-region labels are complete for the selected 20-frame sample; they do not validate the continuous face tracks across the full recordings.

1. Dataset and reproducible processing

The project combines five L-CAS thermal recordings with respiration and heartbeat reference CSVs. The local audit covers 14,154 thermal frames. The archived official ground-truth ZIP matches the local CSV files. Thermal matrices were decoded from the ROS bags, paired to PNGs by message order after count verification, and analyzed with image-header timestamps.

The observed image timing is irregular: median image intervals are about 37-38 ms and effective per-recording rates span 21.8-24.9 Hz. This measured timing was retained rather than assuming a nominal camera rate. Raw mono16 counts were preserved and Celsius was calculated with the dataset encoding formula, (count - 1000) / 10.



Owner-confirmed event map: 0 = no event, 1 = inhalation, 2 = exhalation (not author-verified)

Figure 1. Processing and evaluation path implemented in the repository.

Stage	Implemented	Evidence / note
Input audit	5 ROS bags + official reference CSVs	14,154 frames; local reference files match archived official source
Time alignment	Image-header timestamps	Anestis reference pairing uses nearest time; ordinal pairing is invalid
Temperature	Raw counts + Celsius	Counts preserved; official encoding conversion recorded
Label map	0 / 1 / 2 working map	Owner-confirmed; authors' numeric codebook not located

The official dataset description gives a two-minute recording protocol and the camera encoding; the paper says respiration references were made by manually annotating inhalation times. Neither published description defines the numeric codes in the supplied CSV files. The report therefore treats 0 = no event, 1 = inhalation, 2 = exhalation as a project-owner-confirmed working convention, not an author-verified codebook.

2. Face and facial-region reference review

The project owner completed the selected sample: 20 frames across five subjects, with face and five candidate facial regions reviewed on each frame. There are 115 visible boxes, five regions marked not visible, and no unresolved region decisions. This is a small, single-reviewer sample.

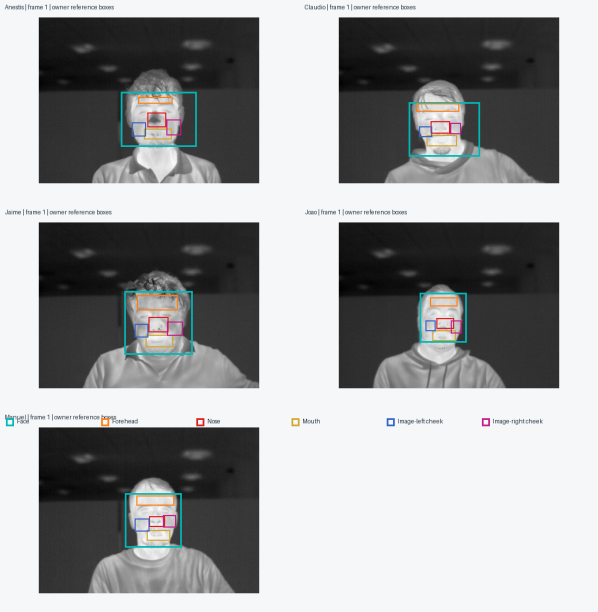


Figure 2. Owner-drawn face and facial-region boxes on one manually reviewed frame per subject. Colors identify the face, forehead, nose, mouth, and image-left/right cheeks.

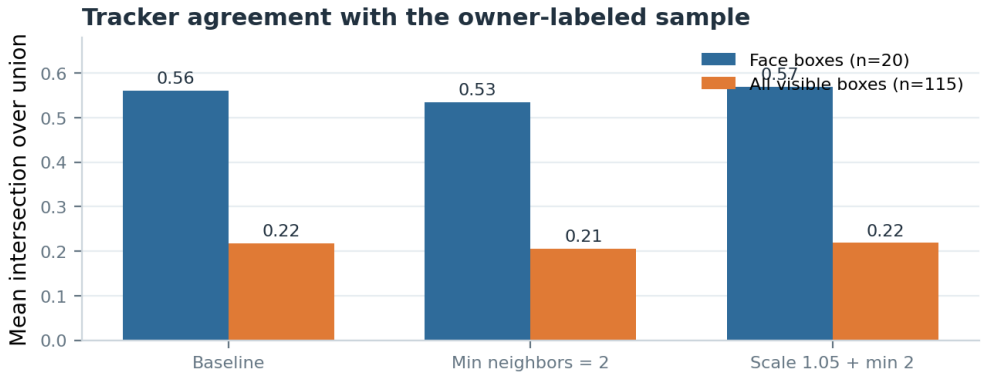


Figure 3. Tracker agreement against all owner-labeled boxes in the 20-frame sample. Face-only IoU is higher than agreement over all facial subregions.

Baseline mean IoU is 0.56 for the 20 face boxes, but 0.22 across all 115 visible face and subregion boxes. The lower-threshold candidates do not improve all-box agreement (0.21 and 0.22). These values describe agreement with one annotator's sparse reference sample; they are not full-recording accuracy.

3. ROI geometry and localization

Subregion agreement exposes a localization gap: the baseline face-box mean IoU is 0.56, while individual facial-region means are about 0.13-0.19. The main constraint is placing a small region inside the correct face area, not simply generating a box for every frame.

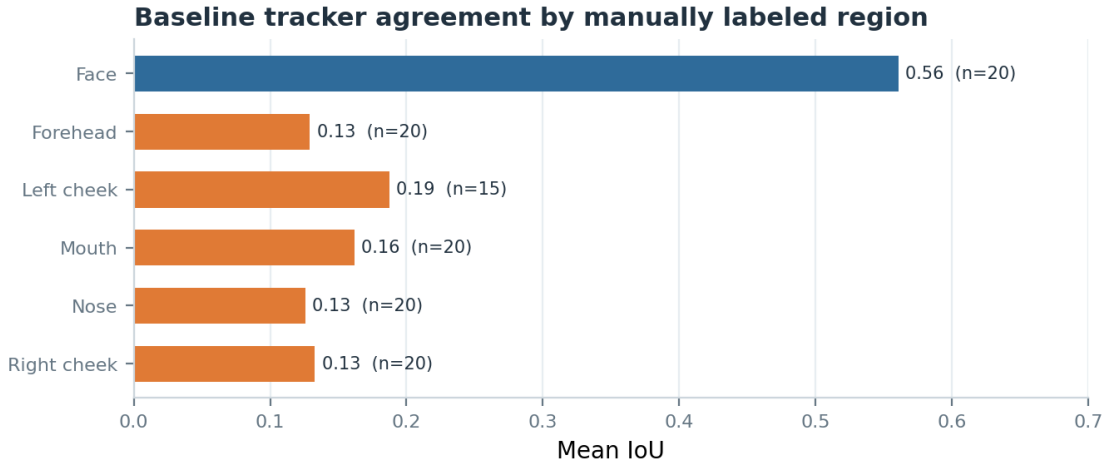


Figure 4. Baseline mean IoU by region; the number beside each bar is the number of visible labeled boxes. The left cheek has 15 labels because five were marked not visible.

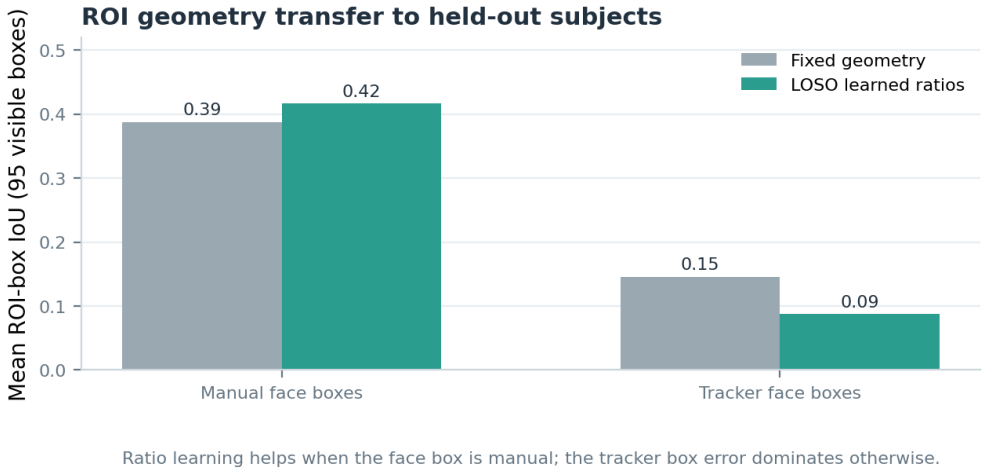


Figure 5. Leave-one-subject-out comparison of fixed geometric ROI ratios with ratios learned on the other four subjects. Manual face boxes isolate geometry; tracked face boxes include detector error.

With manually annotated face boxes, learned ratios raise overall mean IoU from 0.387 to 0.416. With baseline tracker boxes, IoU falls from 0.145 to 0.087. This shows the tracker box can dominate downstream ROI overlap; no ratio scheme is selected for production.

4. Thermal signals and graph filtering

Regional temperatures were extracted from each tracked ROI and plotted against measured time. The example below shows the tracked nose trace and a 10-40 BPM bandpass signal. Large excursions occur during the movement portion, so a full-recording spectral rate can be sensitive to motion and tracking shifts.

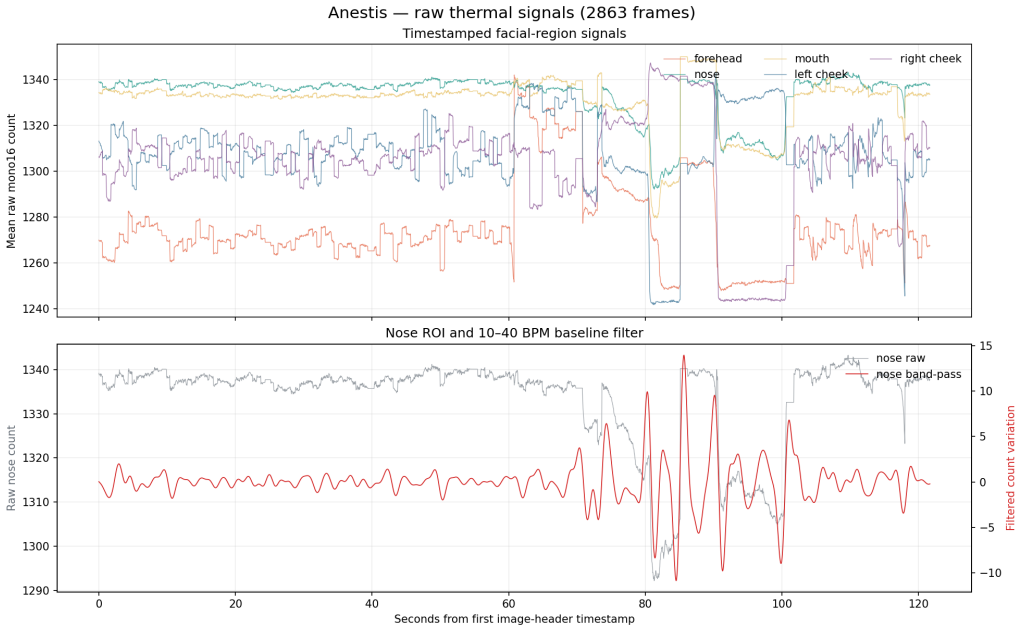


Figure 6. Anestis timestamped raw ROI signals and filtered nose trace. The plot is a diagnostic visualization; it does not by itself demonstrate event accuracy.

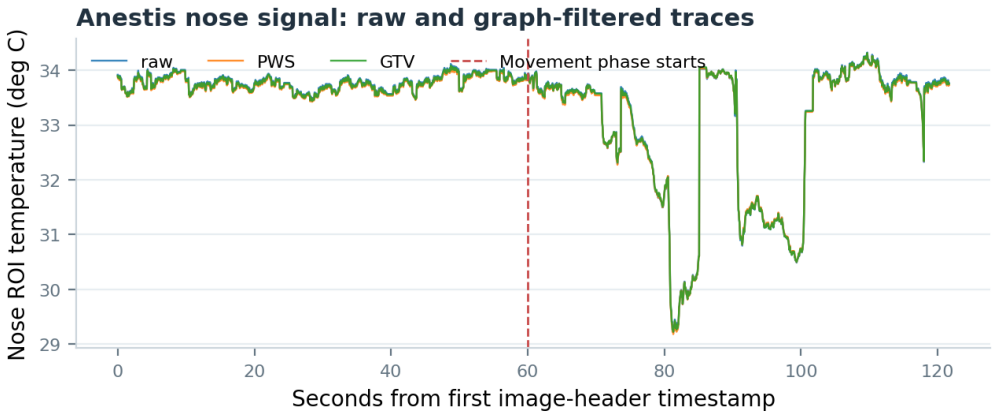


Figure 7. Raw, PWS, and GTV nose traces for Anestis. The tested graph filters make only small changes to the dominant rate estimates.

Signal	LOSO mean absolute rate error
Raw	4.72 BPM
PWS	4.62 BPM
GTV	4.91 BPM

PWS lowers the five-subject LOSO rate MAE by only 0.10 BPM relative to raw (4.72 to 4.62 BPM); GTV is worse at 4.91 BPM. This is not strong evidence that graph filtering improves the baseline.

5. Baseline rates and region ablation

The conventional baseline linearly resamples the measured timestamps, applies a fourth-order zero-phase Butterworth filter in the 10-40 BPM band, and selects the Hann-periodogram peak. Nose-region rates are compared with inhalation-interval rates under the working label map.

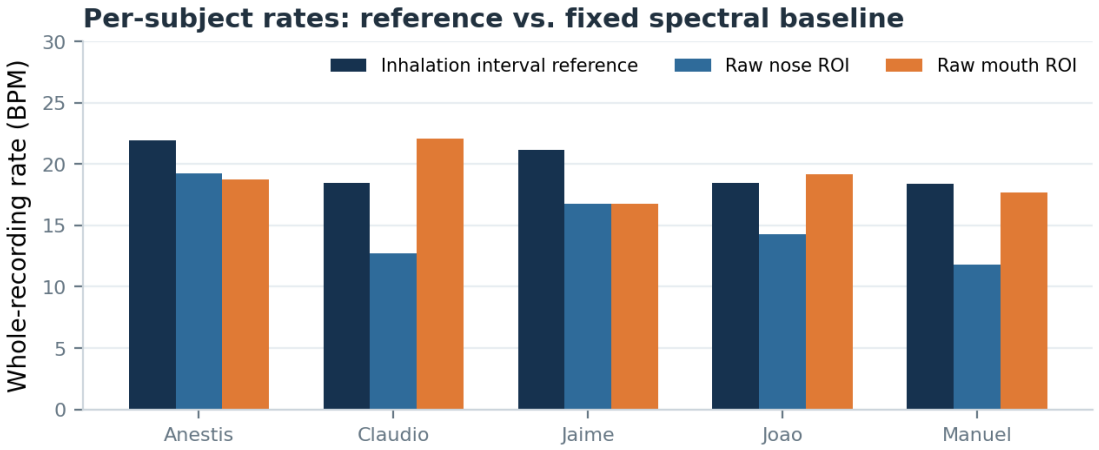


Figure 8. Reference event-span rate and two raw ROI spectral rates by subject. The mouth signal is shown as an exploratory candidate, not a held-out choice.

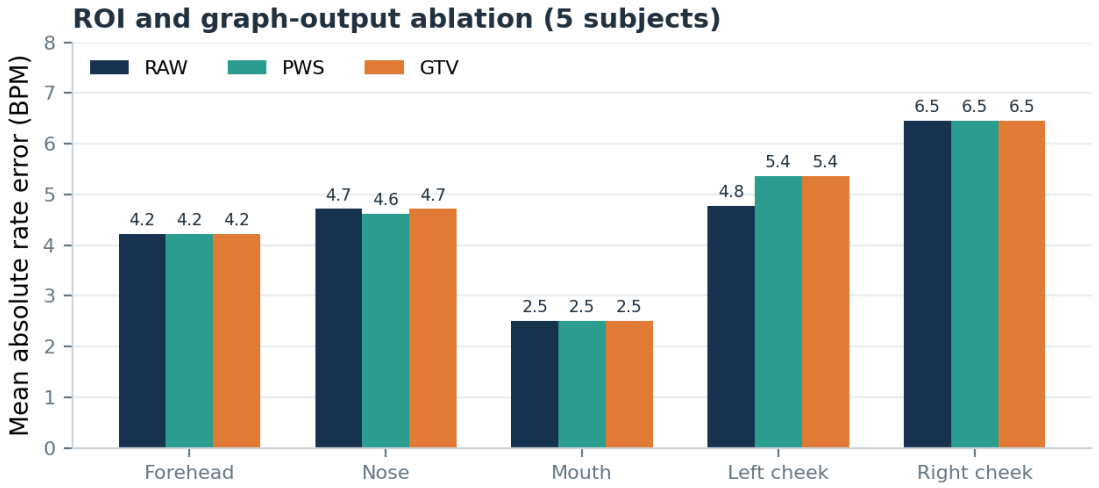


Figure 9. Fixed 10-40 BPM spectral rate error by region and graph output across five subjects. Values are mean absolute errors against event-span rates.

The best observed region in this analysis is mouth/raw at 2.51 BPM MAE, compared with 4.72 BPM for nose/raw (about 47% lower). Because the same five subjects were used to rank the regions, that reduction is exploratory and likely optimistic; it needs confirmation on new subjects.

6. Event-level baseline comparison

The TCN uses raw, PWS, and GTV nose signals to detect sparse inhalation timestamps. Outer leave-one-subject-out evaluation uses inner subject-wise validation for threshold selection. The comparison is against a fourth-order bandpass peak/valley detector with its polarity and prominence selected within the training subjects.

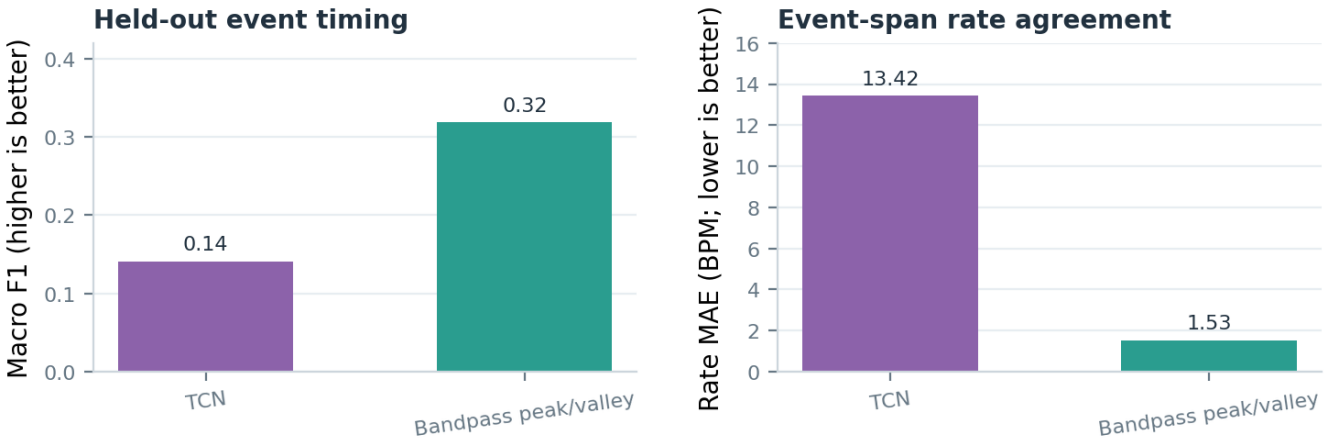


Figure 10. Macro event F1 and event-span rate MAE from held-out subjects. Higher F1 is better; lower rate MAE is better.

Metric	TCN	Bandpass peak/valley
Macro event F1	0.141	0.318
Macro recall	0.083	0.322
Event-span rate MAE	13.42 BPM	1.53 BPM
Median-interval rate MAE	11.97 BPM	2.06 BPM
Count-rate MAE	16.50 BPM	1.10 BPM

Across all five held-out subjects, the TCN identified 16 events and missed 181; the comparator had higher macro F1 but still made substantial timing errors. The comparator's count-rate MAE is only 1.10 BPM because the number of predicted events is close to the number of references. Rate agreement can therefore look good even when event-level F1 is modest.

Interpretation: the current evidence favors the simple bandpass event baseline over the TCN. The TCN is not selected, and neither method is a validated clinical monitor.

7. Conclusions and next steps

1. Data processing is traceable: the local reference CSVs were compared with the archived official source, raw bag frames were decoded, timestamps were retained, and all 14,154 PNG frames were audited.

2. The face and facial-region annotation workflow is complete for the selected 20-frame sample. It contains 115 visible boxes and five explicit not-visible decisions.

3. The owner-labeled sample exposes a real engineering issue: face-box agreement is moderate, while small facial-region overlap is low. Continuous ROI quality remains a limiting factor.

4. The raw mouth-region baseline has the lowest in-sample region-ablation rate error, but the five-subject sample is too small to select a final ROI.

5. The TCN event detector did not beat the bandpass peak/valley baseline under nested subject-wise evaluation. Graph filtering produced no convincing rate improvement over raw signals.

Open scientific requirements

Requirement	Why it remains open
Author-verified 0/1/2 semantics	The published dataset description and paper do not define numeric CSV codes.
Independent ROI review	Current labels have one annotator and cover 20 sampled frames, not full tracks.
Larger held-out evaluation	Only five subjects are available; region ranking on them is exploratory.
Continuous-waveform TCN denoising	The dataset supplies event timestamps, not a clean continuous respiration waveform.

Reproduce the report

From the project root, run `python scripts/build_project_report.py`. The report draws on the processed results under `data/processed/` and the owner review under `data/annotations/`.

Sources

- L-CAS, Thermal Physiological Monitoring Dataset (official dataset description, recording protocol, and mono16 temperature conversion).

- S. Cosar et al., Thermal Camera Based Physiological Monitoring with an Assistive Robot, IEEE EMBC, 2018 (respiration reference based on manually marked inhalation instants).

- C. Yang et al., Graph-Based Denoising for Respiration and Heart Rate Estimation During Sleep in Thermal Video, IEEE Internet of Things Journal, 2022.

- B. Lokendra and G. Puneet, AND-rPPG: A Novel Denoising-rPPG Network for Improving Remote Heart Rate Estimation, Computers in Biology and Medicine, 2022.

- LCAS, official physiological monitoring ROS node (rate estimator and 10-40 BPM search band).

Report generated from local project outputs on 31 October 2026. Reported errors are descriptive research metrics under the project-owner-confirmed working code map; no clinical claims are made.